COLLABORATOR · LEADER · ENGINEER

Kiel Lazarski — **10+ years of medicinal chemistry**, advancing small molecules from hit identification through development candidate nomination

NOW Director, Medicinal Chemistry, Flagship Pioneering · 35 patents and publications · uses top-end AI models to prompt-engineer bespoke software and automate tasks

DEGRADERS

Delivered development candidates for PROTAC and non-covalent programmes. Ternary complex models evaluated by conformational analysis and historic SAR, then used to design PROTACs with improved degradation kinetics.

DESIGN

Structure-based, property-driven design: X-ray, MD, FEP, QSAR. Orally bioavailable, CNS-penetrant molecules outside the Rule of 5. Hits progressed from DEL, virtual screens and fragment libraries.

TEAMS

Leads internal and external chemistry teams across FTE, FFS and CRO models. Mentors scientists and builds high-performing multi-disciplinary teams.

THE EXERCISE

Two tasks — **a past project revisited with Claude**, and a programme plan for a target with real unmet need

ANSWERED WITH the published pan-mutant RET degrader programme, and four tools built for this exercise

TASK 1 — THE PROJECT

The pan-mutant RET degrader programme. Published in full, so every decision behind it can be shown and questioned.

TASK 2 — THE PLAN

Four tools that nominate a target, choose a ligase, measure the interface and propose a compound set.

WHERE THEY MEET

RET is the worked example. The same decisions, taken by hand then and by the tools now.

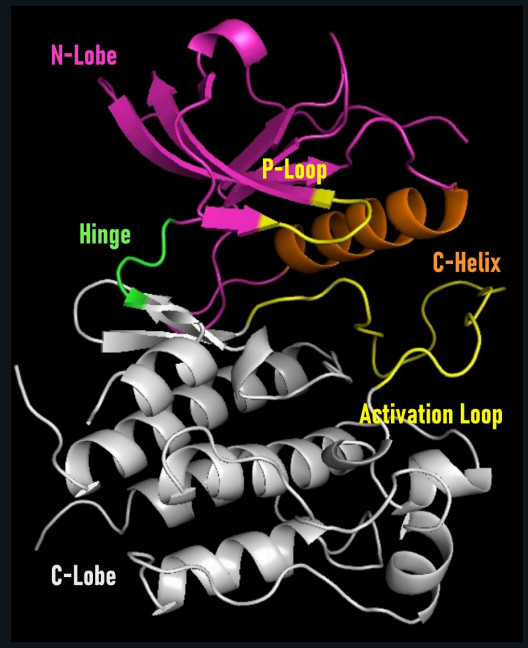
TASK 1 — RET DEGRADER

Choosing where the linker leaves the kinase: **the shortest path to solvent that avoids the moving parts**

The hinge joins the two lobes. It is rigid, short, and faces solvent directly.

Distal to the hinge the linker has to cross the P-loop, the C-helix and the activation loop — the motifs that move when the kinase is regulated.

Hypothesis: the shortest exit that avoids those motifs forms the ternary complex faster and costs the least binary affinity.



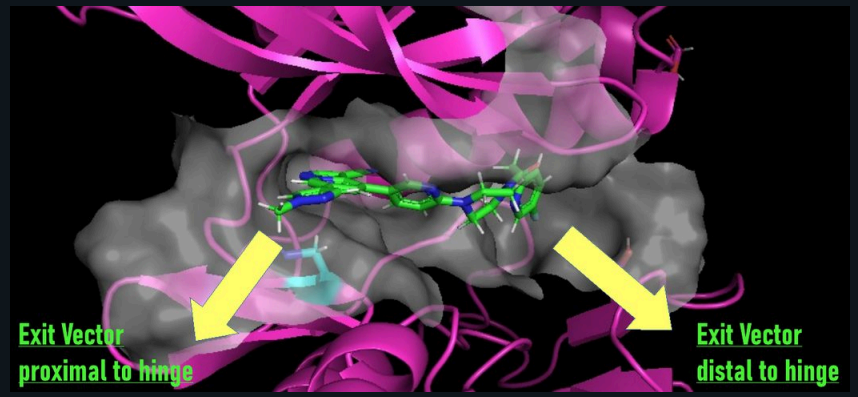
Kinase domain, structural features labelled

Orsi, Lazarski, Fitzgerald, Henderson et al. RSC Med. Chem. 2025, 16, 4781–4795

TASK 1 — RET DEGRADER

Two candidate exit vectors, modelled: **one leaves beside the hinge, the other through the activation loop**

The same warhead in the same ATP site, with two places for the linker to leave.



Side view — monovalent ligand in the ATP site, both candidate exit vectors marked



Top-down view of the modelled ternary complex

Orsi, Lazarski, Fitzgerald, Henderson et al. RSC Med. Chem. 2025, 16, 4781–4795

TASK 1 — TARGET PRODUCT PROFILE

Choosing which mutants the molecule must cover: **G810R is the one the inhibitors miss**

Structural biology, biochemistry, translational biology and in vivo pharmacology read the same literature; biochemical screening and ternary-complex modelling then set the priority.

G810R sits on the exit vector itself — the highest risk and the highest reward in the profile.

SELPERCATINIB	IC_{50}, ATP = K_m	IC_{50}, ATP = 30 MM	K_m (MM)
RET wild type	1 nM	14 nM	—
V804L gatekeeper	2 nM	—	0.07
V804M gatekeeper	2 nM	24 nM	0.44
A883F C-lobe	4 nM	—	—
M918T C-lobe	2 nM	—	0.45
G810R exit vector	—	**530 nM**	—



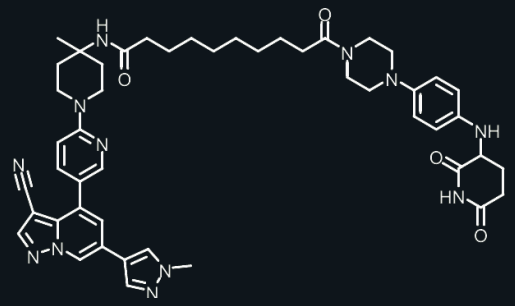
Selpercatinib docked into the wild-type RET structure, mutation sites marked

Orsi, Lazarski, Fitzgerald, Henderson et al. RSC Med. Chem. 2025, 16, 4781–4795

TASK 1 — RET DEGRADER

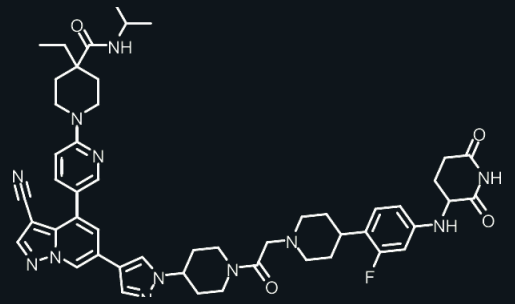
Proximal beat distal in cells: **21 nM DC_{50} against no measurable degradation**

Same warhead, same ligase, same linker chemistry. Only the attachment point moved. Emax is protein remaining, so lower is better.



DISTAL TO HINGE

NA / 185 nM IP / 61% Emax



PROXIMAL TO HINGE

21 nM DC_{50} / 20 nM IP / 24% Emax



Both candidate vectors on the modelled complex

Orsi, Lazarski, Fitzgerald, Henderson et al. RSC Med. Chem. 2025, 16, 4781–4795

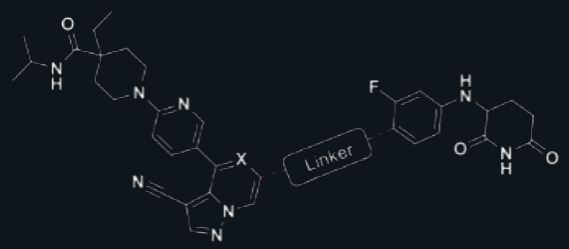
TASK 1 — PUBLISHED SERIES

The published series starts at compound 15: **potent on RET, with no oral exposure**

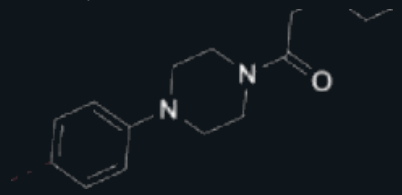
Point mutations and fusions both drive constitutive activation, so degradation offers pan-mutant and pan-fusion pharmacology with better kinome selectivity.

G810R is where second-line inhibitors lose potency.

VEGFR2 is the high-homology anti-target.



LINKER, COMPOUND 15 X = C



CPD	RET KD DC_{50} / EMAX	G810R DC_{50} / EMAX	RET-WT IC_{50}	PVEGFR2 IC_{50} / EMAX	CL ML/MIN/KG	PO AUC NG·H/ML	F
15	5 nM / 23%	103 nM / 37%	43.6 nM	>10 µM / 61%	—	—	—

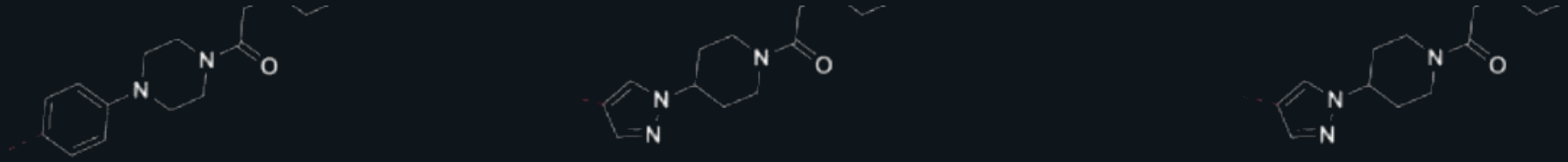
Orsi, Lazarski, Fitzgerald, Henderson et al. RSC Med. Chem. 2025, 16, 4781–4795

TASK 1 — PUBLISHED SERIES

Two changes, two problems: **the exit-vector angle, then the basicity of the linker**

15 → 16. Narrowing the exit-vector angle from 180° to 144° improved binary binding, and degradation with it.

16 → 17. Lowering basicity and buried surface area cut clearance and raised permeability: bioavailability moved from 0.9% to 18%.



Linkers of compounds 15, 16 and 17

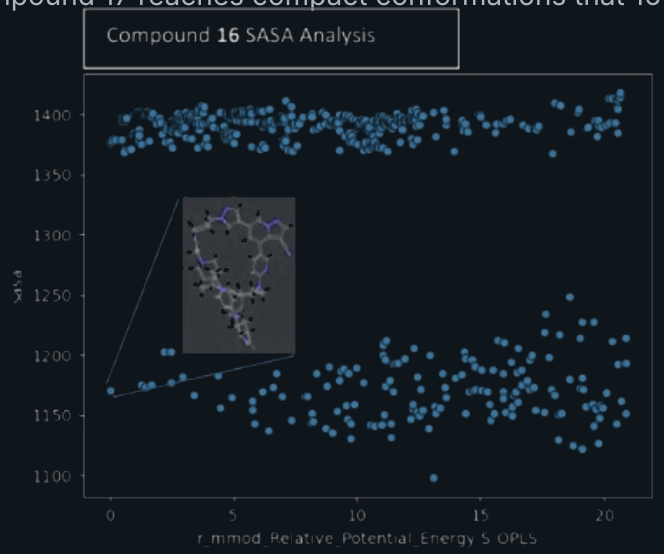
CPD	RET KD DC_{50} / EMAX	G810R DC_{50} / EMAX	RET-WT IC_{50}	PVEGFR2 IC_{50} / EMAX	CL ML/MIN/KG	PO AUC NG·H/ML	F
15	5 nM / 23%	103 nM / 37%	43.6 nM	>10 µM / 61%	—	—	—
16	1.9 nM / 13%	24 nM / 24%	9 nM	1162 nM / 30%	31	8.09	0.88%
17	1.8 nM / 13%	19 nM / 21%	3.5 nM	671 nM / 17%	16	181	18%

Orsi, Lazarski, Fitzgerald, Henderson et al. RSC Med. Chem. 2025, 16, 4781–4795

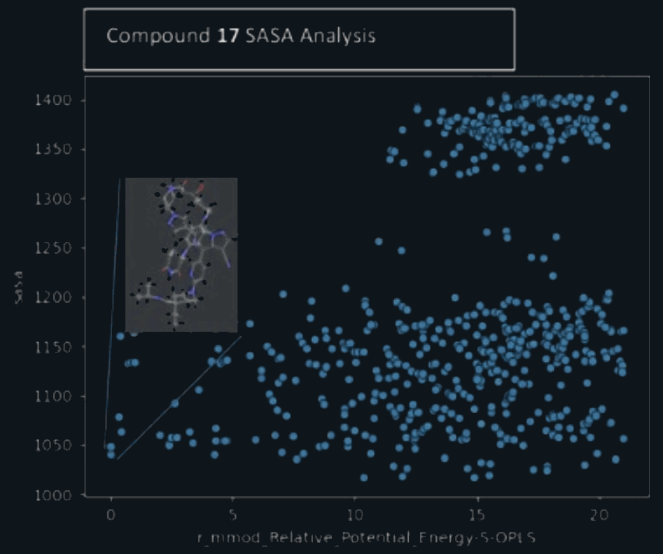
TASK 1 — PUBLISHED SERIES

Why the property change worked: **the molecule folds in on itself once the solvent is gone**

Conformers from a molecular dynamics ensemble, plotted as solvent-accessible surface area against relative energy. Compound 17 reaches compact conformations that 16 does not.



Compound 16



Compound 17 — a low-SASA population appears

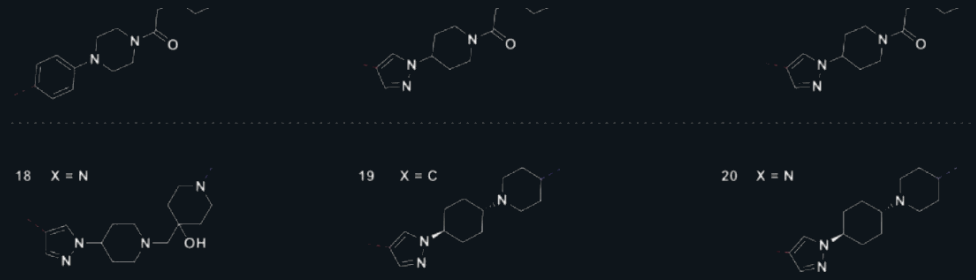
Orsi, Lazarski, Fitzgerald, Henderson et al. RSC Med. Chem. 2025, 16, 4781–4795

TASK 1 — PUBLISHED SERIES

Linker length and rigidity trade against the anti-target: **rigidifying recovered VEGFR2 selectivity**

17 → 18. Cutting polar surface area raised permeability again, but VEGFR2 selectivity fell.

18 → 20. A rigid linker recovered it, at 39% oral bioavailability.



Linkers of compounds 15 to 20

CPD	RET KD DC_{50} / EMAX	G810R DC_{50} / EMAX	RET-WT IC_{50}	PVEGFR2 IC_{50} / EMAX	CL ML/MIN/KG	PO AUC NG·H/ML	F
15	5 nM / 23%	103 nM / 37%	43.6 nM	>10 µM / 61%	—	—	—
16	1.9 nM / 13%	24 nM / 24%	9 nM	1162 nM / 30%	31	8.09	0.88%
17	1.8 nM / 13%	19 nM / 21%	3.5 nM	671 nM / 17%	16	181	18%
18	1.4 nM / 11%	19 nM / 24%	—	660 nM / 18%	15	601	53%
19	1.6 nM / 8%	19 nM / 21%	1.7 nM	>2670 nM / 43%	12	325	23%
20	1.7 nM / 8%	21 nM / 22%	4.25 nM	5141 nM / 42%	18	319	39%

Orsi, Lazarski, Fitzgerald, Henderson et al. RSC Med. Chem. 2025, 16, 4781–4795

TASK 1 — WHAT IT TURNED ON

What the programme actually turned on: **three decisions taken before any compound was made**

AND NOW each of those three decisions is one of the tools in the second half of this deck

WHICH MUTANTS

A cross-functional reading of the literature, then biochemistry and ternary-complex modelling, put G810R in the target profile.

WHERE THE LINKER LEAVES

Topology and dynamics ranked the exit vectors. The shortest clear path degraded best; 180° to 144° improved it further.

WHAT SETS EXPOSURE

Linker basicity set clearance. Conformational collapse, predicted by dynamics, set permeability.

TASK 2 · TARGET NOMINATION

Tool 1 — **database-mining web tool** that nominates target–ligase pairs in immunology, inflammation and cardiometabolic disease

BUILT AS tpd_calibration_set.csv · Proteins of Interest page · Glue Site Scan · dock.mjs

WHY IT EXISTS

Outside oncology there is no essentiality screen to borrow, and a degrader is a three-body problem — the ligase cannot be chosen late.

WHAT IT DRAWS ON

Human genetics, Open Targets, GTEx, STRING, UniProt, ChEMBL and the trial registries, plus structures for both partners. All public.

WHAT COMES OUT

A ranked shortlist of target–ligase pairs, each with an evidence tier and a likely modality.

TASK 2 · OMNI TOOL

Tool 2 — **structural workbench** that measures whether a target–ligase interface can host a small molecule

BUILT AS Omni Tool · Exit Vectors page, TC Feasibility page, Cross-Species page

WHY IT EXISTS

Geometric impossibility is usually found after the chemistry. This needs only the two structures.

WHAT IT DRAWS ON

PDB co-crystals and predicted models. Runs offline in a browser — no server, no queue.

WHAT COMES OUT

Where a linker can attach, the reach it needs, and which compounds are impossible.

TASK 2 · GLUE FORGE

Tool 3 — **design tool** that turns a modelled interface into a purchasable compound set

BUILT AS Glue Forge · glue_library_100.xlsx · chemlib.py

WHY IT EXISTS

Every molecular glue so far was found by screening, not designed.

WHAT IT DRAWS ON

The interface map, known ligase-binder series, and make-on-demand catalogues.

WHAT COMES OUT

About a hundred compounds in hypothesis series — a virtual screen against the interface, ranked and ready to order.

TASK 2 — THE WORKFLOW

Each tool narrows what the next one sees

01 Target Nomination

IN
Public databases, expression and structures

OUT
Target–ligase pairs, with an interface

02 Omni Tool

IN
Two structures

OUT
What is geometrically possible

03 Glue Forge

IN
The interface map

OUT
A virtual screen of ~100 compounds

Running. Calibration set, ligase scan and benchmark, all three workbench pages, compound library.

Specified. Nomination scoring function; simulation triage before ordering.

Not started. Pose ranking, co-expression, scaffolding corpus, one pair end to end.

WHAT THE EXERCISE TAUGHT ME

Three things I did not expect: **the data, the cost of software, and how the work divides**

ENDPOINTS DO NOT TRANSFER

Oncology lends an essentiality screen. Inflammation and cardiometabolic disease have none, so evidence was mined by hand — and the weights that rank it are a team decision.

BESPOKE SOFTWARE IS NOW CHEAP

This editor took an hour, at a rounding error of what PowerPoint cost. Software keeps moving toward the single question it is built for.

AGENTS RUN IN PARALLEL

Work divides across parallel agents: research, implementation, and a check that never saw the work it graded.

CONCLUSIONS

What the programme turned on, **and what the tools now do about it**

IN SHORT the decisions that set the RET programme were structural and were taken early; the tools take them earlier and against more candidates

FROM TASK 1

RET turned on three decisions made before any compound: which mutants, where the linker leaves, and what sets exposure.

FROM TASK 2

Each is now a tool that runs on public structures and public data, before chemistry is committed.

STILL OPEN

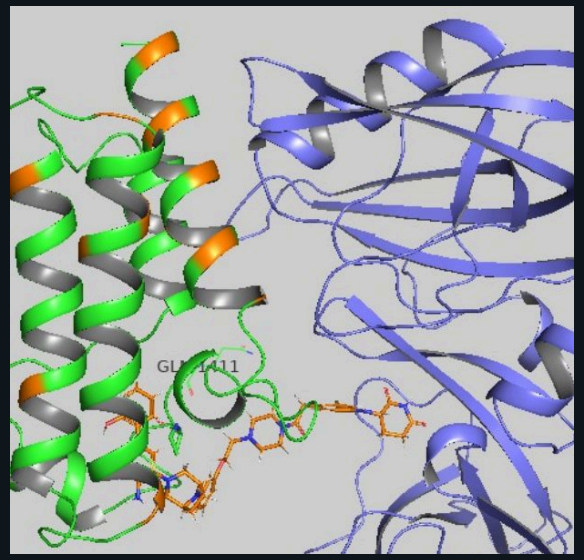
No pair has been run end to end, and the ligase docking benchmark has not yet produced a near-native pose.

SUPPLEMENTAL — SMARCA2

A second degrader programme where selectivity had to come from **the complex, not the warhead**

The SMARCA2 and SMARCA4 bromodomain sites are nearly identical, so a selective binder was not available. Synthetic lethality in SMARCA4-deficient cancers makes the distinction essential.

Models were built by mutating both surfaces to alanine, aligning the sites within linker reach, filtering on HDX signal, then restoring side chains and discarding clashes.



SMARCA2 bromodomain with cereblon

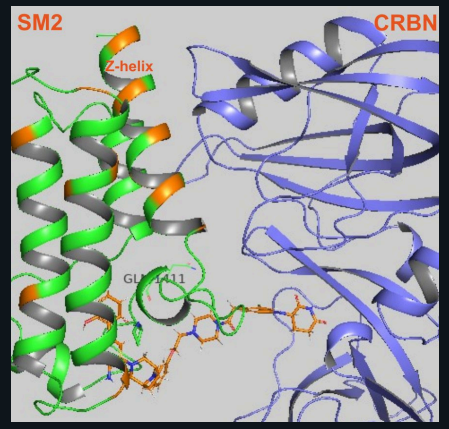
SUPPLEMENTAL — SMARCA2

Choosing between models by the conformations they imply: **prior SAR decided it**

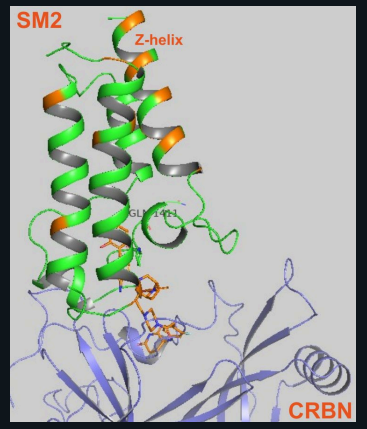
Models were aligned first on the target and then on the ligase, and every point where the proposed bioactive conformations diverged was catalogued.

Three compounds built to favour or block the extended conformation agreed with model 1689.

CPD	WHAT IT TESTS	SMARCA2 DC_{50} / EMAX
001	Steric clash favours extended B	30 nM / 29%
002	Steric clash disfavours extended B	915 nM / 43%
003	Electron repulsion favours extended B	45 nM / 13%



Model 1689 — the differential Z-helix sits in the interface



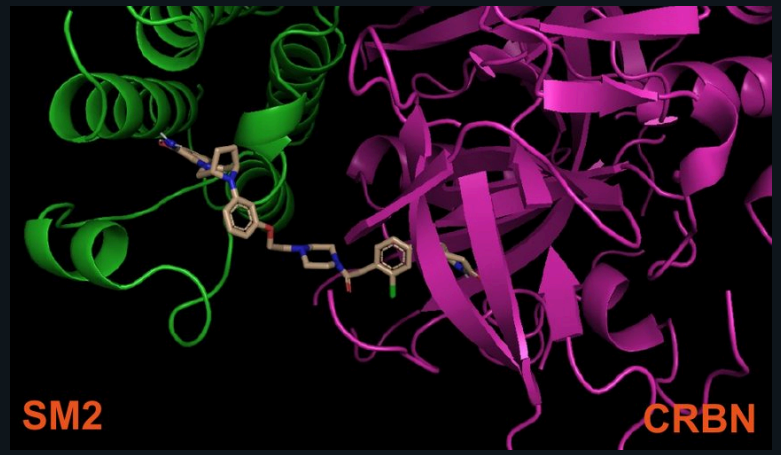
Model 254601

SUPPLEMENTAL — SMARCA2

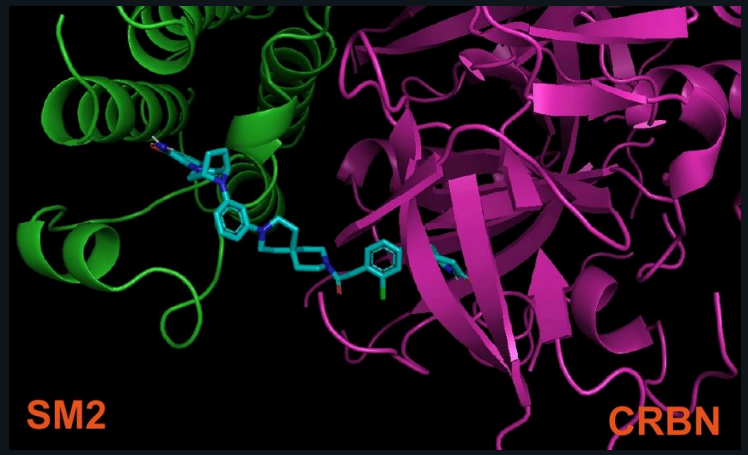
The model used as a design hypothesis: **a spirocycle that pre-organises the linker**

A 6,6-spirocycle was proposed to hold the linker in the modelled conformation without touching either binding site.

Compound 005 kept the potency, improved SMARCA4 selectivity, and cut clearance from 130 to 18 mL/min/kg.



004 — acyclic ethyl ether linker · 40 nM / 23% · CL 130 mL/min/kg



005 — rigid spirocyclic linker · 50 nM / 15% · CL 18 mL/min/kg